



# Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
- M07: LLM Fundamentals
- M08: Prompt Engineering
- M09: Fine-Tuning & PEFT
- M10: RAG & Real-World Projects



# Advanced NLP & Large Language Models (LLMs)

**5.1 Introduction to Applied NLP Tasks**

**5.2 Core Task: Text Classification**

**5.3 Understanding User Intent: Sentiment Analysis**

**5.4 Reducing Information Overload: Text Summarization**

**5.5 Information Retrieval: Question Answering Systems**

**5.6 Task Comparison and the Future of Applied NLP**



# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

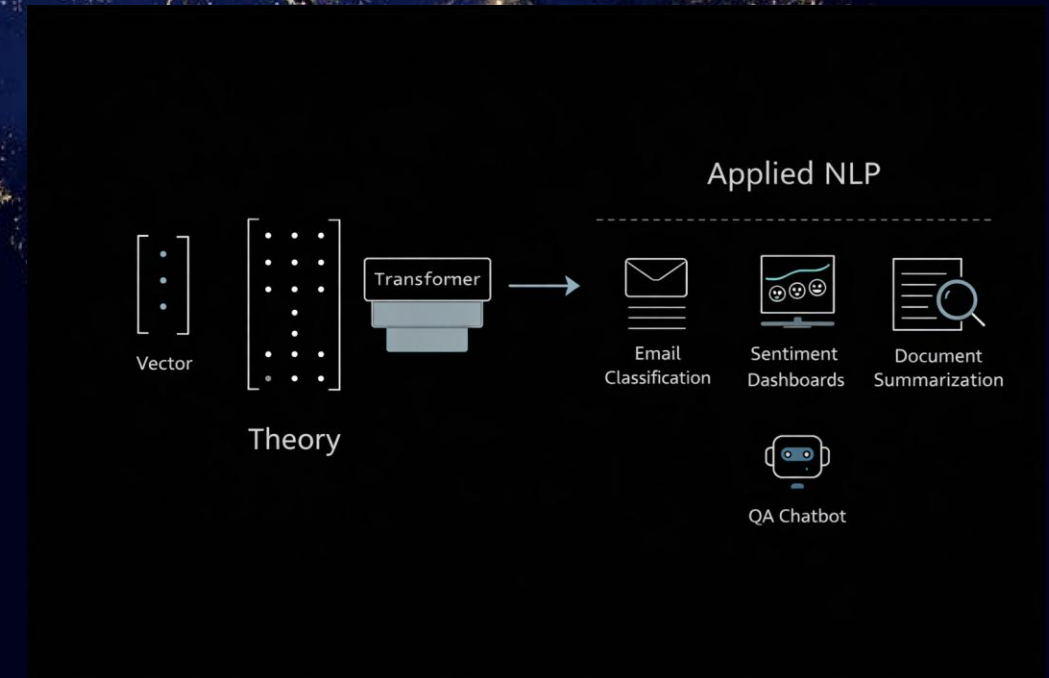
### From Theory to Applied NLP

NLP moves from mathematical modeling to real-world problem solving

Applied NLP focuses on business, industrial, and societal value

Models are evaluated by usefulness, not only benchmark scores

This module connects Transformers to production systems



# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

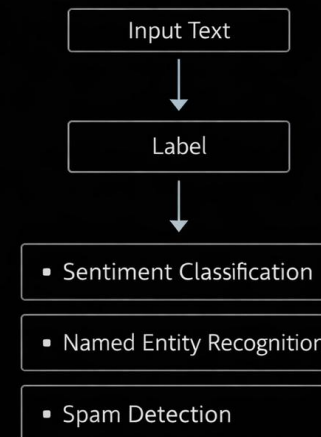
### Applied NLP Task Families

Applied NLP tasks fall into two paradigms

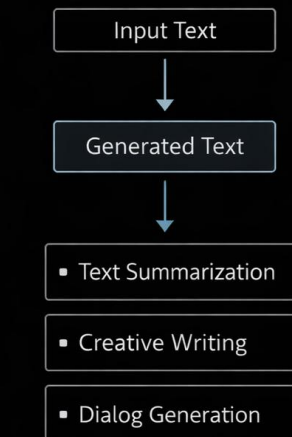
Tasks are defined by what the model produces

Choice impacts architecture, evaluation, and cost

#### Discriminative Tasks



#### Generative Tasks

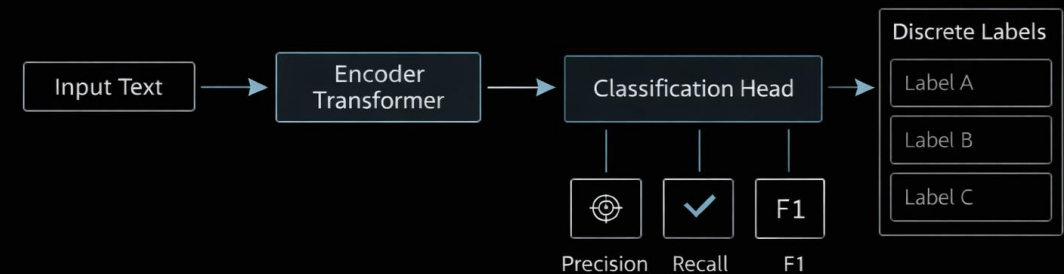


# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

### Discriminative Tasks

- Output: predefined labels
- Goal: decision, filtering, categorization
- Typical architectures: Encoder-only (BERT-style)
- Metrics: Accuracy, Precision, Recall, F1-score
- Examples: classification, sentiment, NER





# Generative Tasks

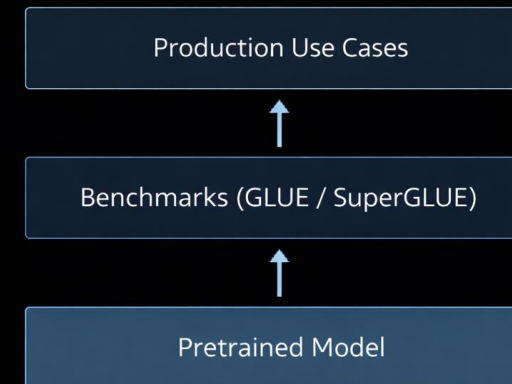
- 
- Diagram illustrating a sequence-to-sequence model architecture:
- Input Context** (represented by a box) feeds into the **Decoder Transformer** (represented by a box).
  - The **Decoder Transformer** outputs a sequence of tokens: "The model can generate relevant ...".
  - The output sequence is shown within a box with arrows indicating **Autoregressive atrows between generated tokens**.
  - The output sequence is followed by an ellipsis (**...**).
  - Below the output sequence, a list of qualities is provided:
    - Fluency
    - Faithfulness
    - Diversity

# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

### Benchmarks and Model Readiness

- Benchmarks standardize model comparison
- GLUE and SuperGLUE measure reasoning ability
- High benchmark scores  $\neq$  production readiness
- Fine-tuning adapts models to domain-specific tasks



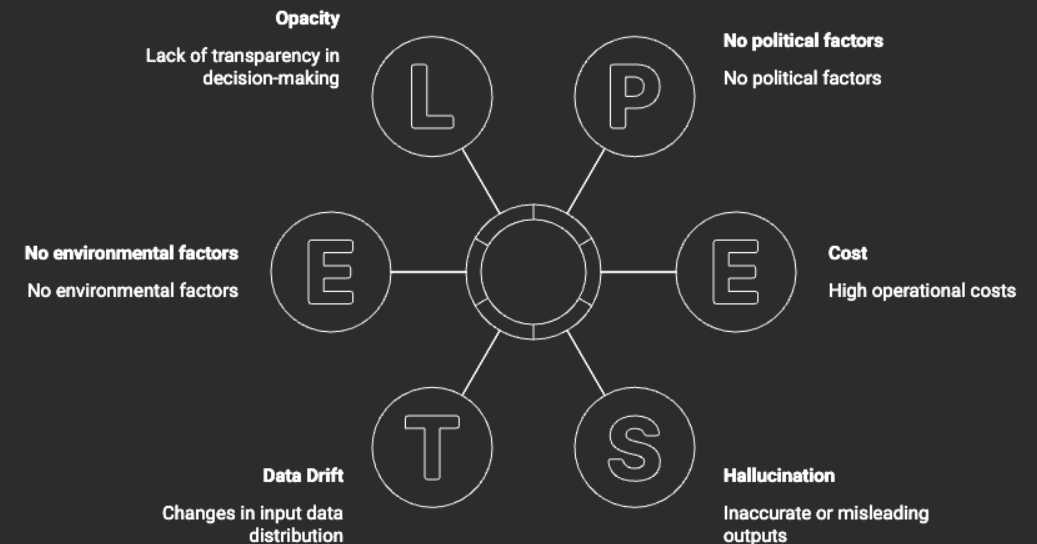
# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

### Production Challenges in Applied NLP

- Benchmarks standardize model comparison
- GLUE and SuperGLUE measure reasoning ability
- High benchmark scores  $\neq$  production readiness
- Fine-tuning adapts models to domain-specific tasks

#### Production NLP System Risks





# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

### Modular Pipelines vs End-to-End Models

- End-to-End: simpler, faster to prototype
- Modular pipelines: more explainable and controllable
- Enterprises often combine both approaches
- Design choice depends on scale and regulation

#### Monolithic vs. Modular LLM Systems

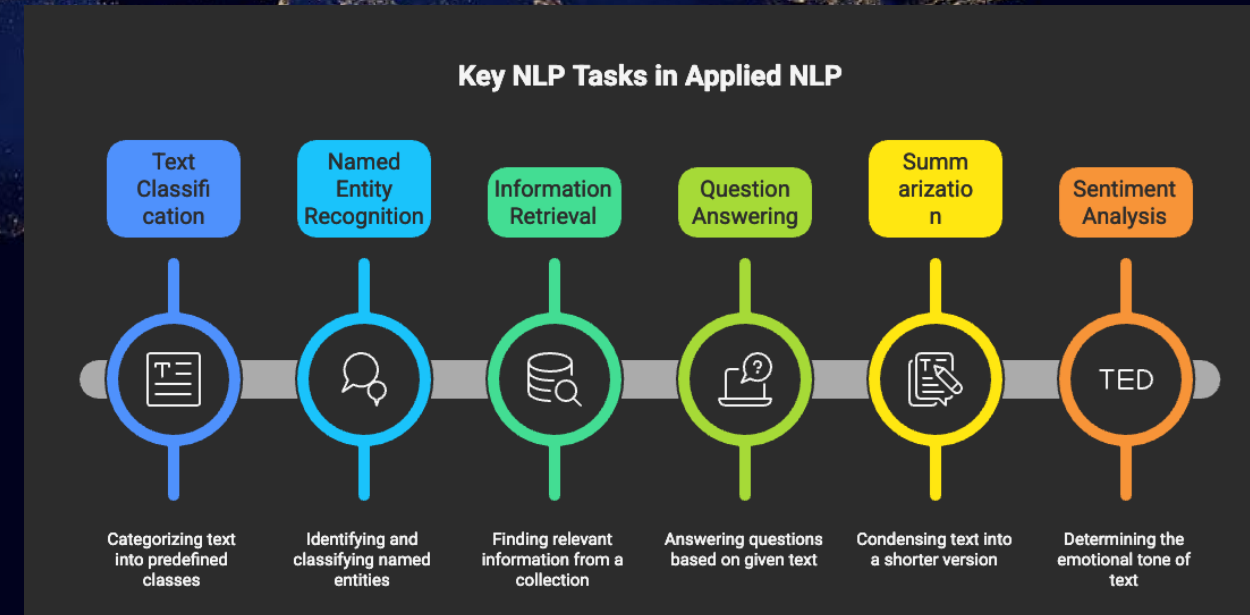
| Characteristic  | Monolithic LLM                       | Modular NLP Pipeline                               |
|-----------------|--------------------------------------|----------------------------------------------------|
| Structure       | Single large rectangular box         | Multiple rectangular boxes connected left-to-right |
| Internal Detail | Minimal, just the label              | NER, Classifier, Retriever, Generator              |
| Flow            | No arrows needed                     | NER → Classifier → Retriever → Generator           |
| Comparison      | Contrasts with multi-module pipeline | Contrasts with single-box system                   |
| Style           | Academic / engineering blueprint     | Academic / engineering blueprint                   |
| Palette         | Monochrome (white and gray) on black | Monochrome (white and gray) on black               |
| Accent Color    | Optional (light blue or teal)        | Optional (light blue or teal)                      |
| Typography      | Sans-serif, neutral                  | Sans-serif, neutral                                |
| Design          | Flat design                          | Flat design                                        |
| Alignment       | Horizontally aligned                 | Horizontally aligned                               |
| Intent          | Visual comparison                    | Visual comparison                                  |

# M05:Applied NLP Tasks

## Chapter 5.1 Introduction to Applied NLP Tasks

### Why This Module Matters

- Applied NLP turns language into action
- Tasks define how AI creates value
- Understanding task structure prevents misuse
- Next chapters deep-dive into each task





# Advanced NLP & Large Language Models (LLMs)

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**5.2 Core Task: Text Classification**

**5.3 Understanding User Intent: Sentiment Analysis**

**5.4 Reducing Information Overload: Text Summarization**

**5.5 Information Retrieval: Question Answering Systems**

**5.6 Task Comparison and the Future of Applied NLP**

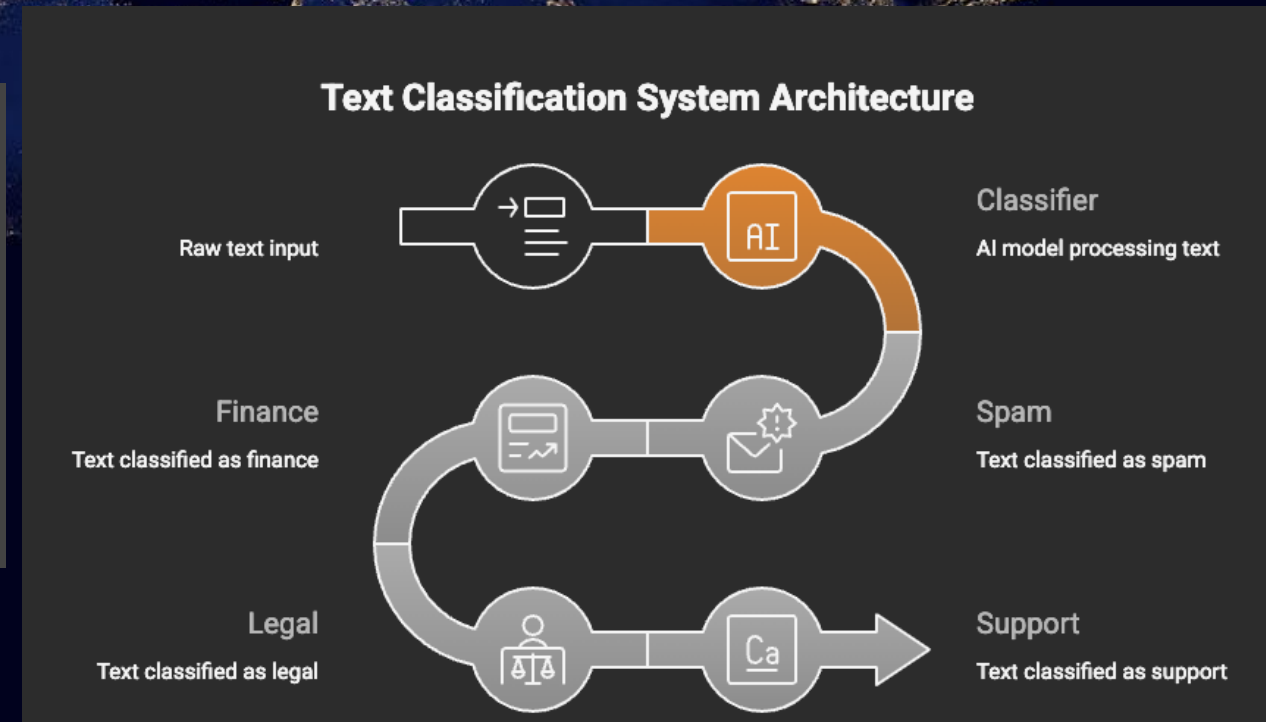


# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### What is Text Classification?

- Task: assign predefined labels to text
- One of the oldest and most deployed NLP tasks
- Foundation for moderation, routing, filtering, analytics
- Used in almost every production NLP system



# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Classification Problem Types

- Binary: two exclusive classes (Yes / No)
- Multi-class: one label from many
- Multi-label: multiple labels per document
- Correct framing defines model architecture

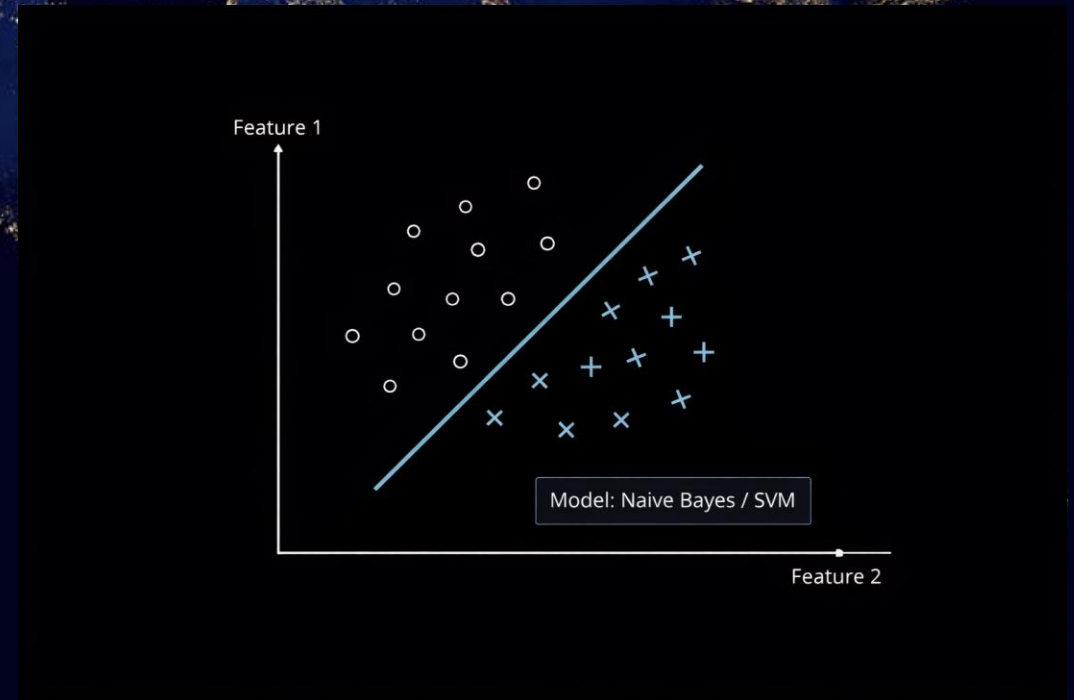
| Classification Paradigms   |                                                                                                  |                                                                                                    |                                                                                           |                                                                                           |
|----------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Characteristic             |  Output Boxes |  Active Outputs |  Arrow |  Title |
| Binary Classification      | Two                                                                                              | One                                                                                                | Single                                                                                    | Binary Classification                                                                     |
| Multi-class Classification | Multiple                                                                                         | One                                                                                                | To set                                                                                    | Multi-class Classification                                                                |
| Multi-label Classification | Multiple                                                                                         | Several                                                                                            | To set                                                                                    | Multi-label Classification                                                                |

# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Traditional Classification Methods

- Based on word frequency statistics
- Require manual feature engineering
- Still useful for small or structured datasets
- Serve as strong baselines



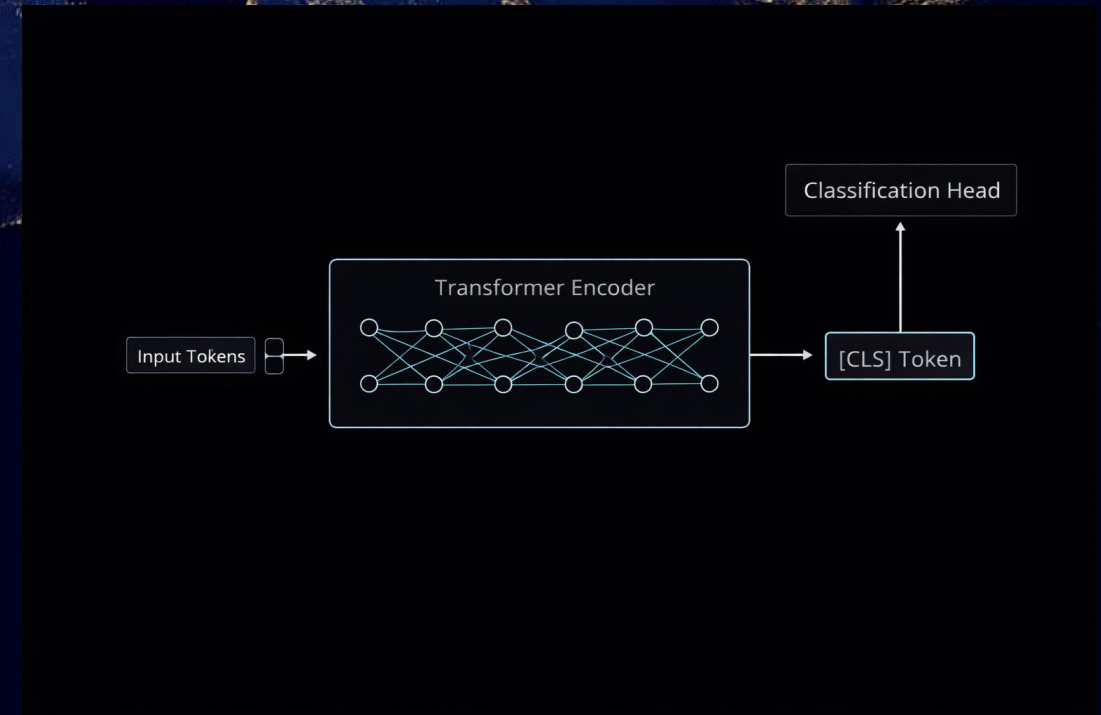


# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Deep Learning Classification

- Models learn features automatically
- Capture semantic meaning, not just keywords
- Transformers are the modern standard
- Context-aware classification



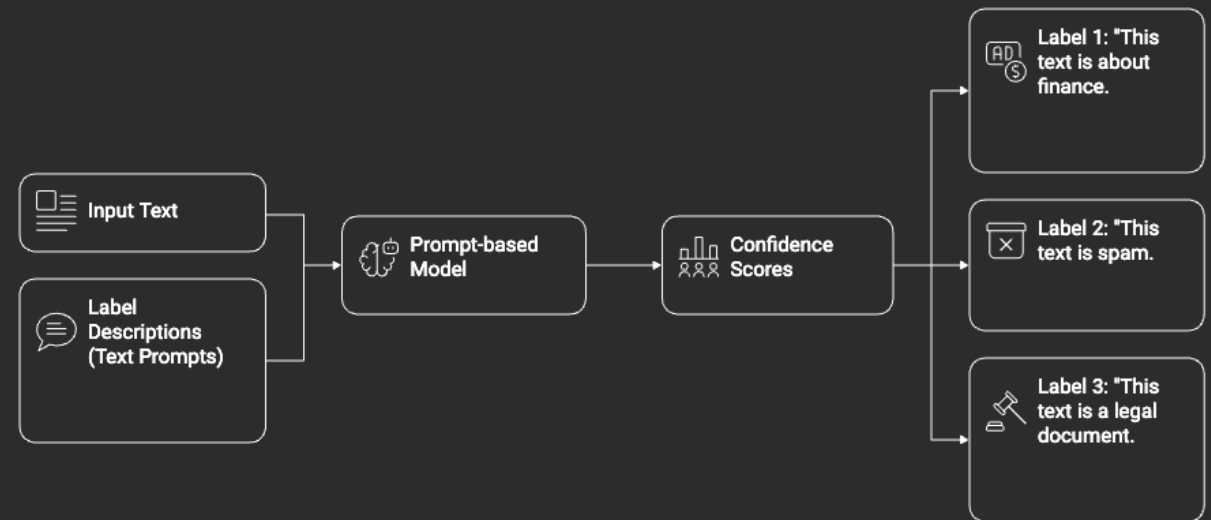
# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Zero-Shot Classification

- No task-specific training required
- Uses natural language label descriptions
- Extremely fast to deploy
- Ideal for dynamic taxonomies

#### Prompt-Based Classification System

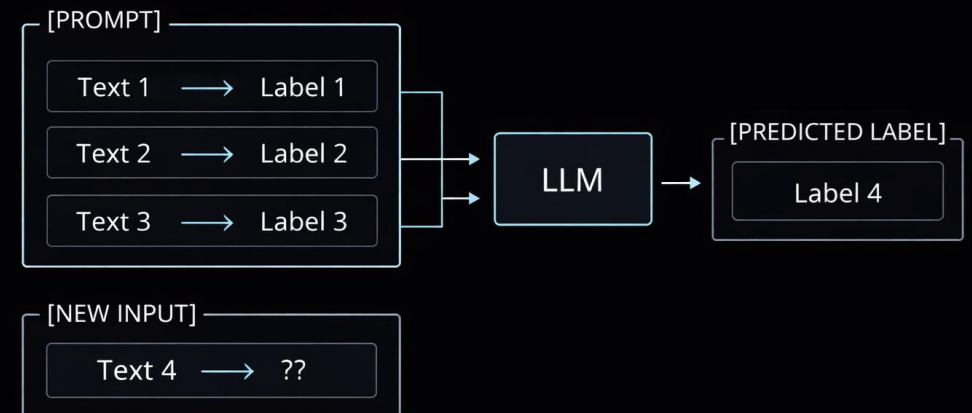


# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Few-Shot Classification

- Uses a few labeled examples in the prompt
- Improves boundary definition
- Reduces annotation cost dramatically
- Common in enterprise rapid prototyping





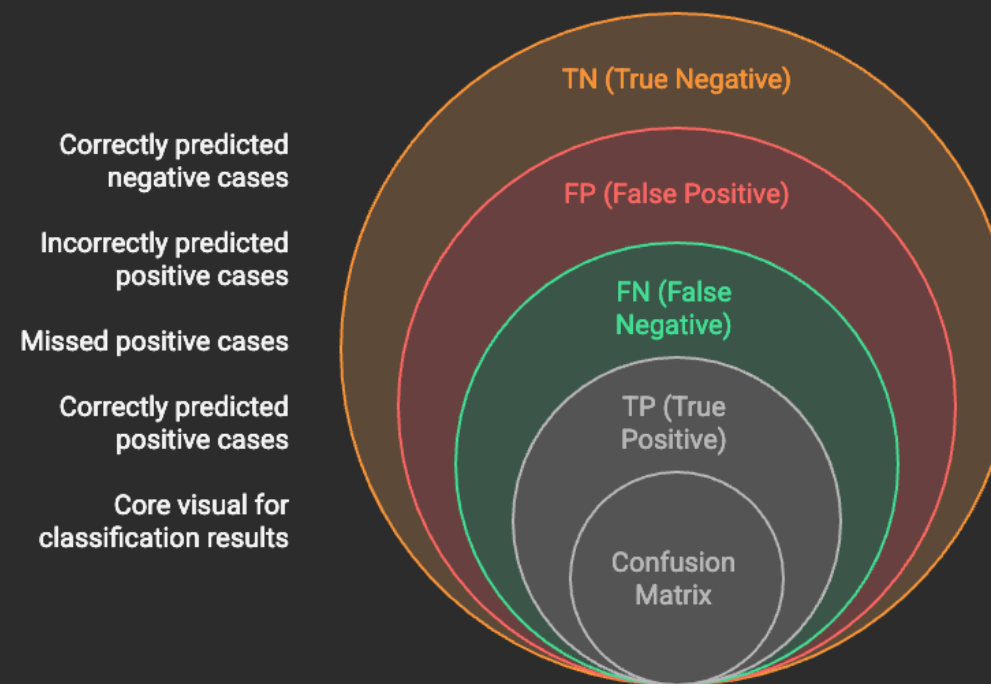
# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Evaluation Metrics

- Accuracy is often misleading
- Precision controls false positives
- Recall controls false negatives
- F1-score balances both
- Metric choice depends on risk

#### Confusion Matrix and Evaluation Metrics

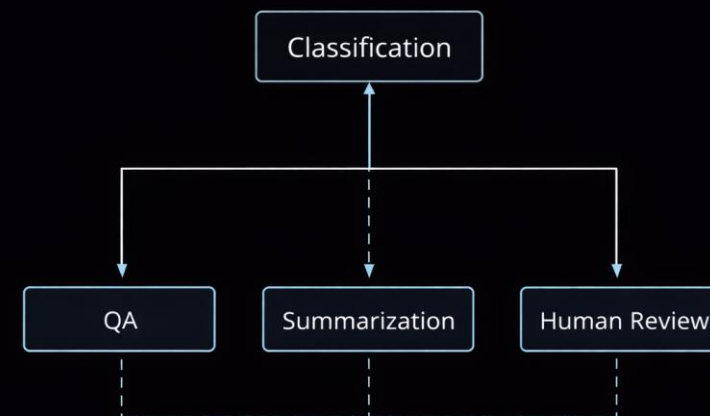


# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Text Classification in Production

- First decision layer in many pipelines
- Often precedes routing, retrieval, or QA
- Must handle drift and class imbalance
- Strong impact on downstream tasks

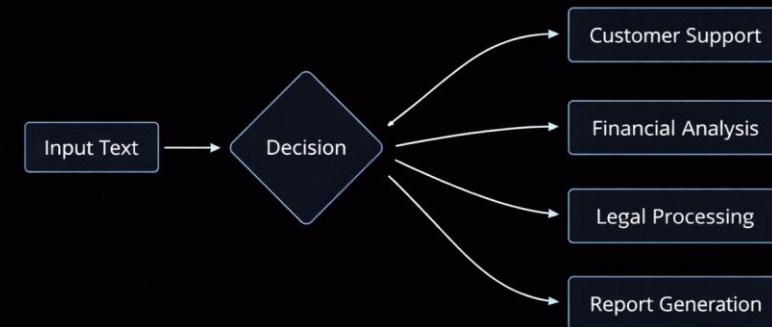


# M05:Applied NLP Tasks

## Chapter 5.2 – Text Classification

### Key Takeaways

- Classification transforms language into decisions
- Architecture choice matters more than model size
- Zero-shot enables rapid deployment
- Metrics define real-world success







# Advanced NLP & Large Language Models (LLMs)

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5.2 Core Task: Text Classification

5.3 Understanding User Intent: Sentiment Analysis

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5.5 Information Retrieval: Question Answering Systems

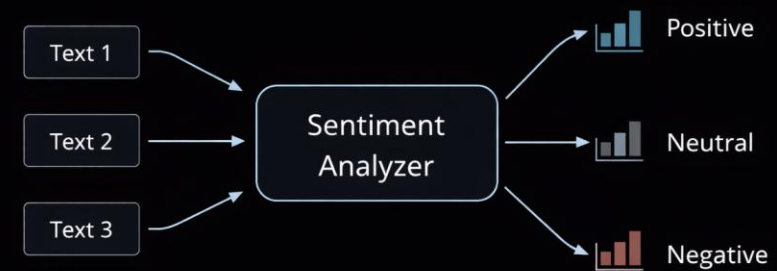
5.6 Task Comparison and the Future of Applied NLP

# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### What is Sentiment Analysis?

- Task: detect subjective intent in text
- Focuses on opinions, emotions, attitudes
- Specialized form of text classification
- Core signal for user and market intelligence



# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Document-Level Sentiment

- One sentiment for the entire document
- Simple and computationally efficient
- Works well for short reviews or posts
- Loses internal nuance

#### Document to Sentiment Score Mapping

Characteristic



Layout



Structure



Style



Intent

Description

Black background,  
document block on  
left, sentiment score  
gauge on right

Left-to-right flow:  
Document →  
Sentiment Score  
Gauge

Clean dashboard  
aesthetic,  
monochrome  
palette, flat design

Visualize document  
to score mapping,  
emphasize  
simplicity



# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Aspect-Based Sentiment Analysis (ABSA)

- Sentiment linked to specific entities or features
- Enables fine-grained insight
- Essential for product and brand analysis
- Industry standard for actionable feedback

#### Aspect-Based Sentiment Analysis

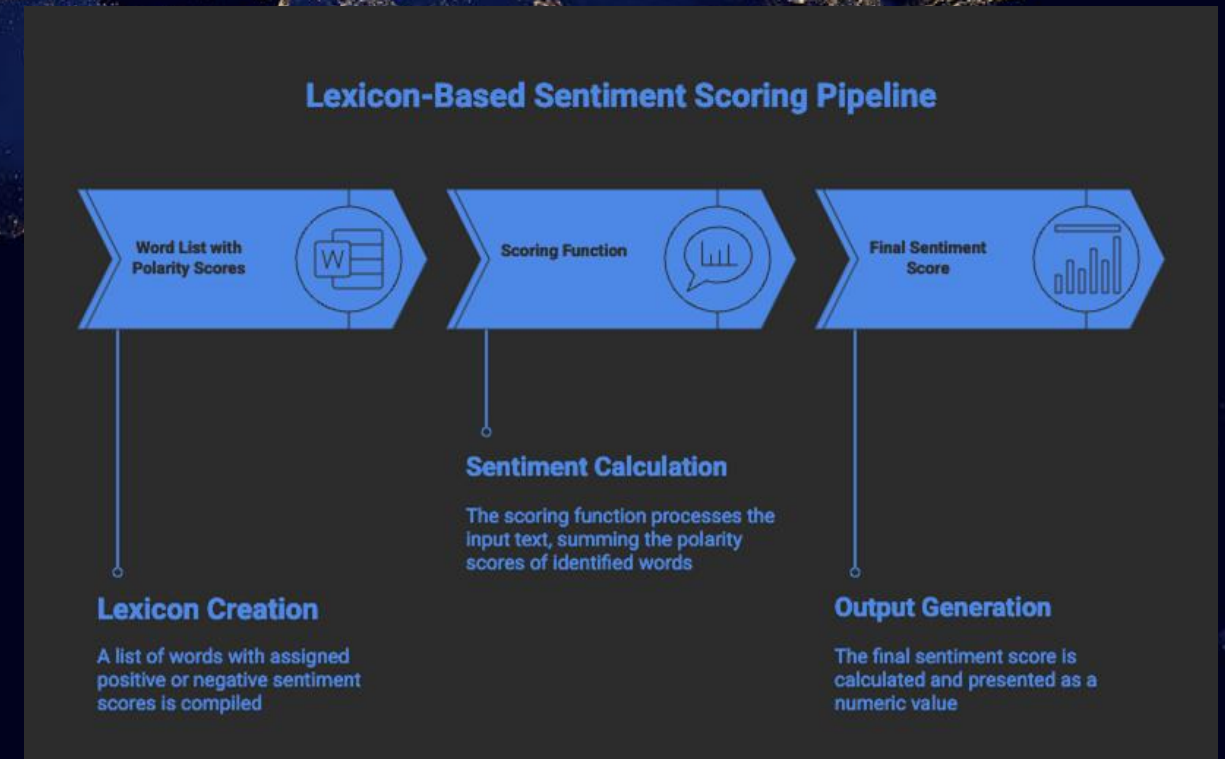
| Characteristic     | Description                                           |
|--------------------|-------------------------------------------------------|
| Layout             | Sentence in center, aspect nodes around               |
| Polarity Outputs   | Arrow from aspect to sentiment label                  |
| Structure          | Sentence is central reference, aspects branch         |
| Color-Coded Arrows | Blue for positive, gray for neutral, red for negative |
| Style              | Technical NLP annotation, flat design                 |
| Intent             | Visualize aspect-based sentiment analysis             |

# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Lexicon-Based Approaches

- Use predefined polarity dictionaries
- Fully explainable
- Fast and lightweight
- Fail under context, irony, or negation

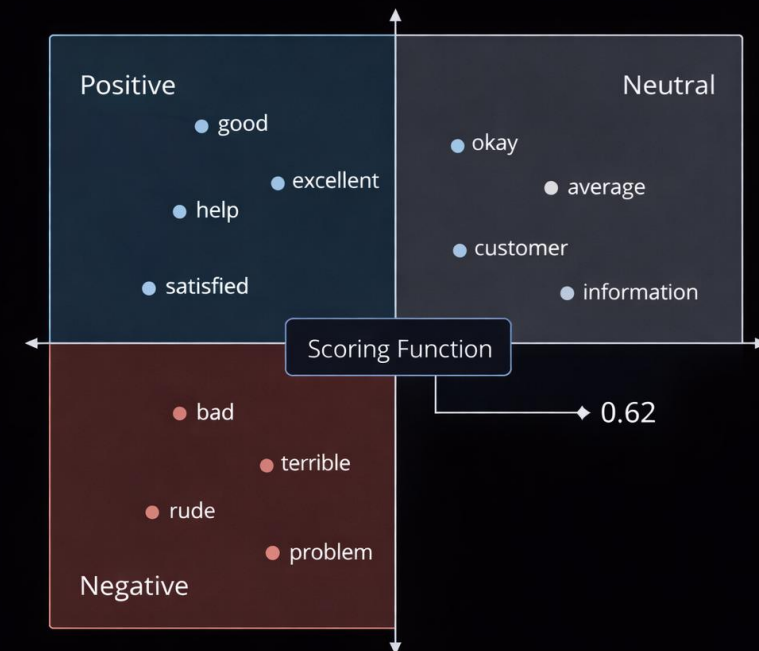


# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Embedding-Based Sentiment Models

- Meaning represented in latent vector space
- Generalizes beyond exact keywords
- Robust to synonyms and paraphrases
- Foundation for transformer-based sentiment





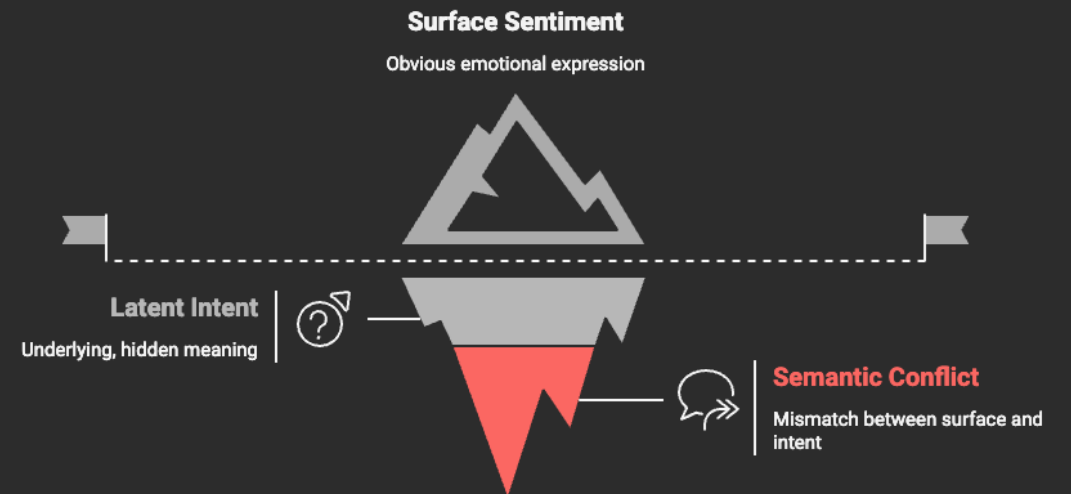
# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Negation and Sarcasm Challenges

- Negation flips polarity meaning
- Sarcasm contradicts surface wording
- Requires long-range context
- Transformers outperform classical models

#### Semantic Conflict: Surface Sentiment vs. Latent Intent

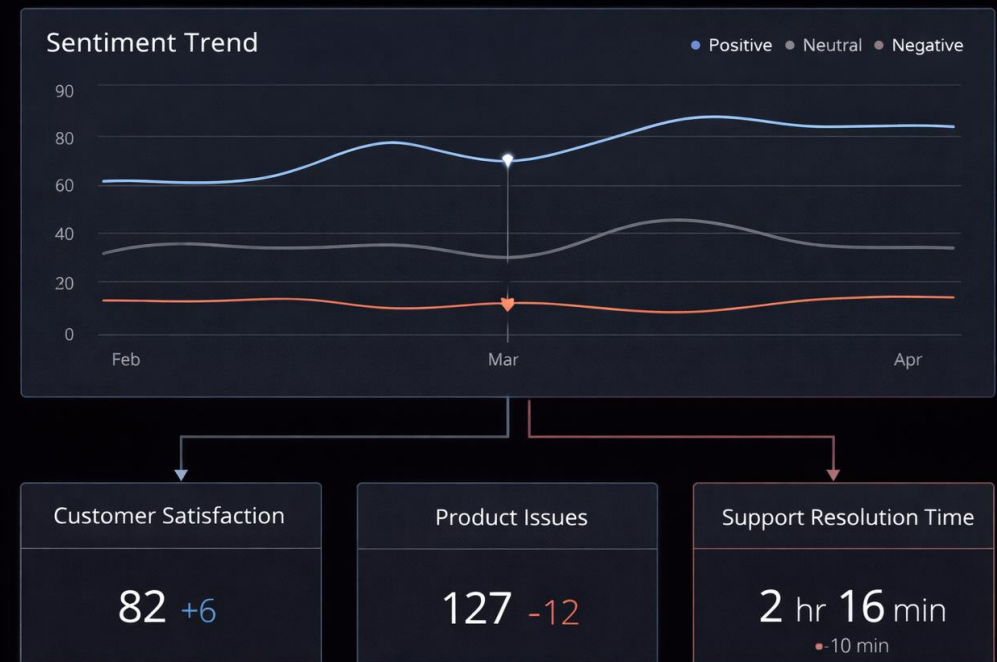


# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Sentiment in Market Intelligence

- Tracks public perception at scale
- Correlates with sales and reputation
- Enables early crisis detection
- Drives automated prioritization

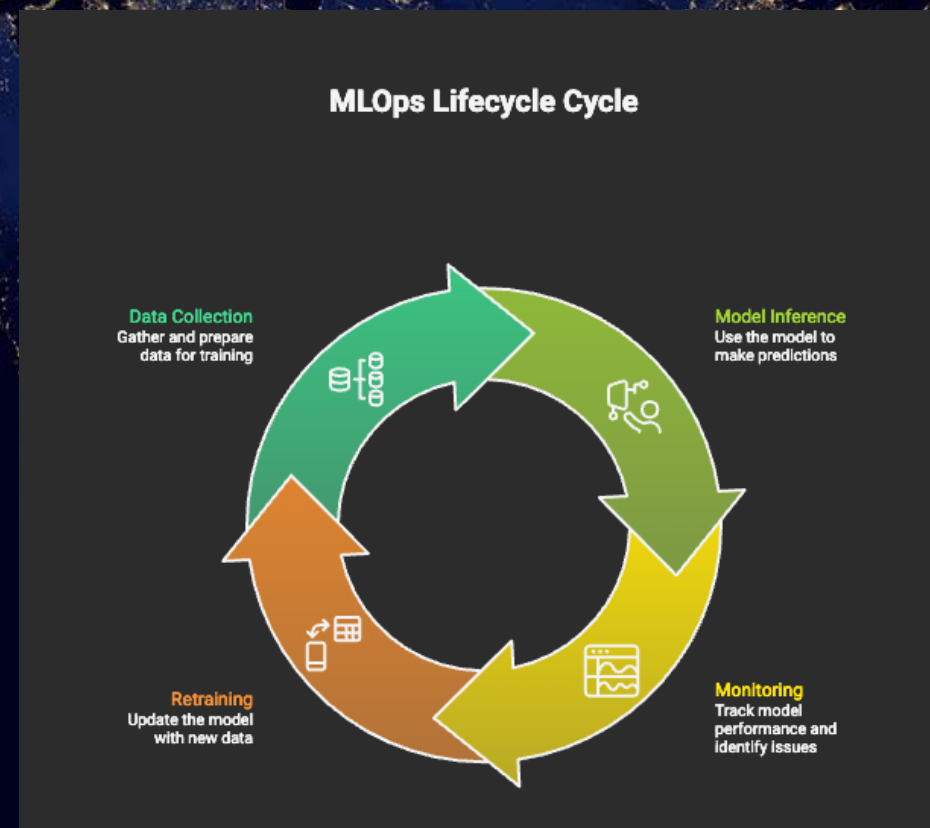


# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Operational Considerations

- Domain-specific language matters
- Models degrade with cultural drift
- Requires periodic re-evaluation
- Sentiment  $\neq$  truth



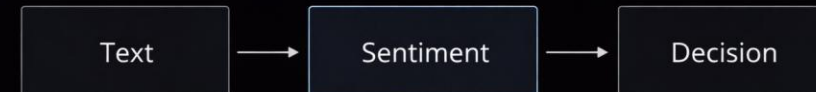


# M05:Applied NLP Tasks

## Chapter 5.3 – Understanding User Intent: Sentiment Analysis

### Key Takeaways

- Sentiment reveals user intent
- Aspect-level analysis enables action
- Context is the main technical challenge
- Critical signal for applied NLP systems





# Advanced NLP & Large Language Models (LLMs)

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5.6 Task Comparison and the Future of Applied NLP

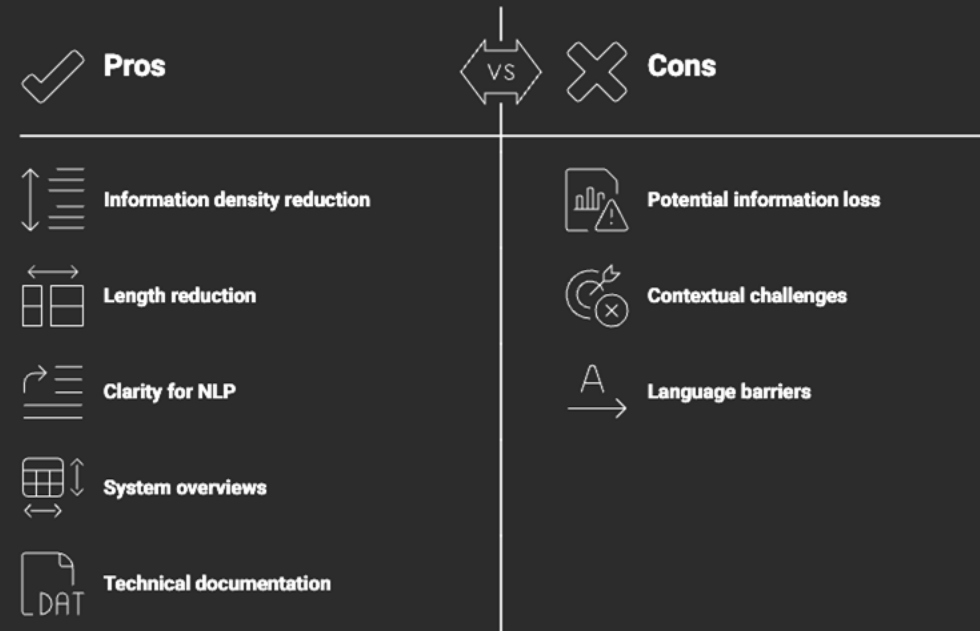
# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### What is Text Summarization?

- Task: compress long text into key information
- Preserves meaning while reducing length
- Essential for scale and decision speed
- Widely used in legal, news, intelligence

#### Document summarization





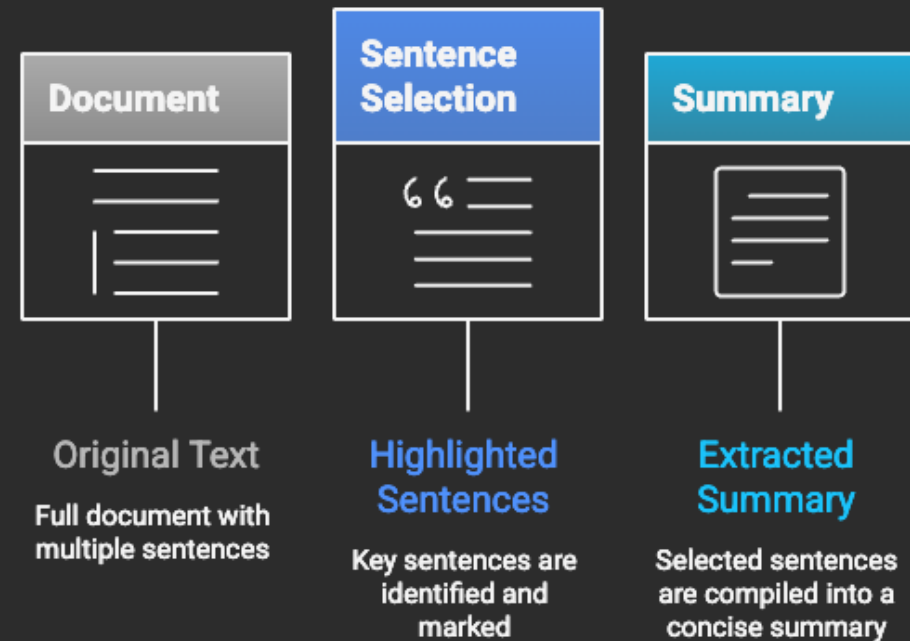
# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Extractive Summarization

- Selects existing sentences
- High factual accuracy
- No text generation
- Often less fluent

#### Extractive Summarization Process

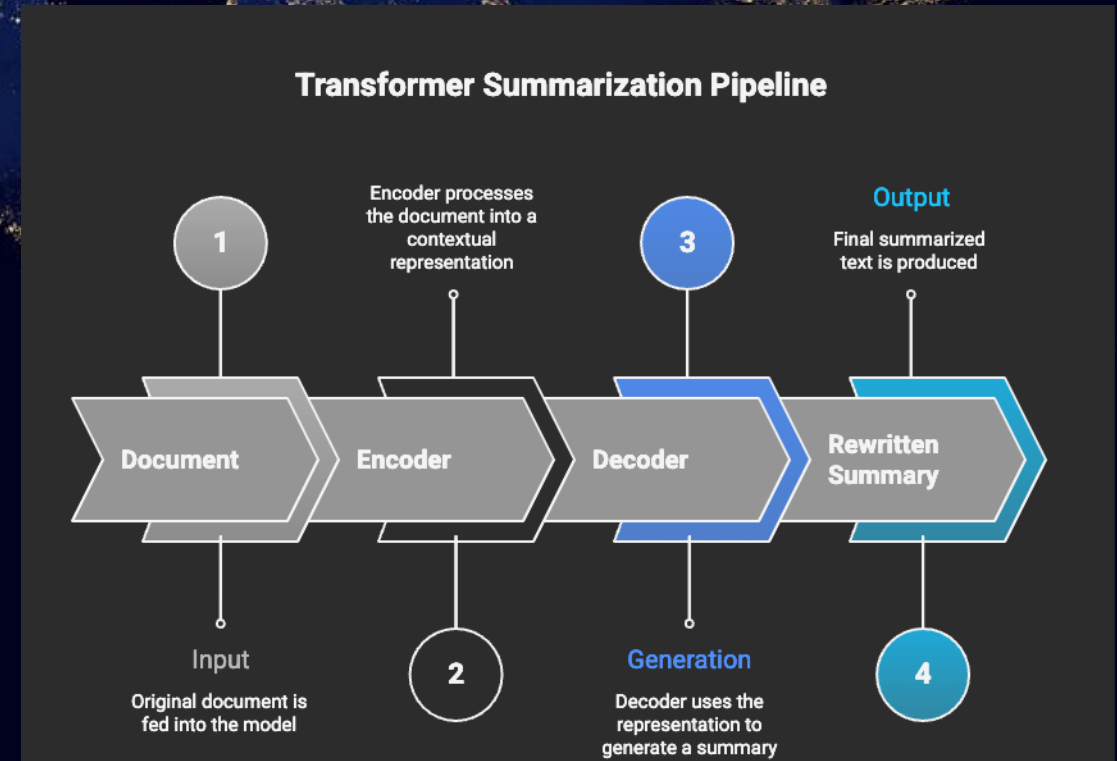


# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Abstractive Summarization

- Generates new sentences
- Higher information density
- More natural language
- Risk of hallucination

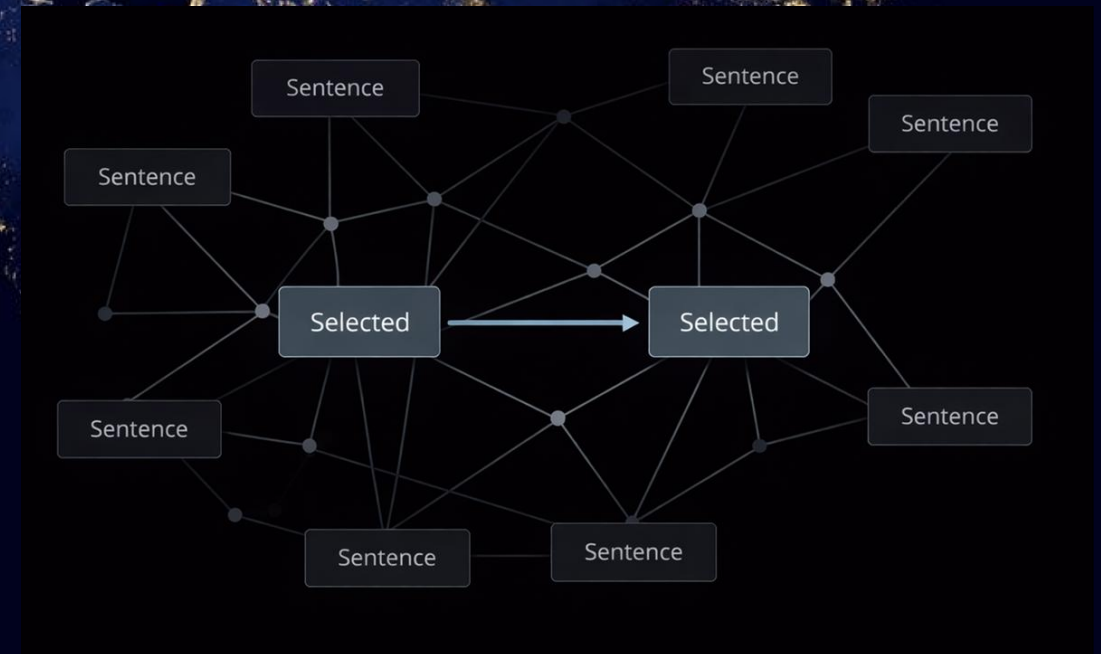


# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Traditional Extractive Techniques

- Graph-based ranking (TextRank)
- Sentence similarity scoring
- No deep semantics
- Still useful for formal text



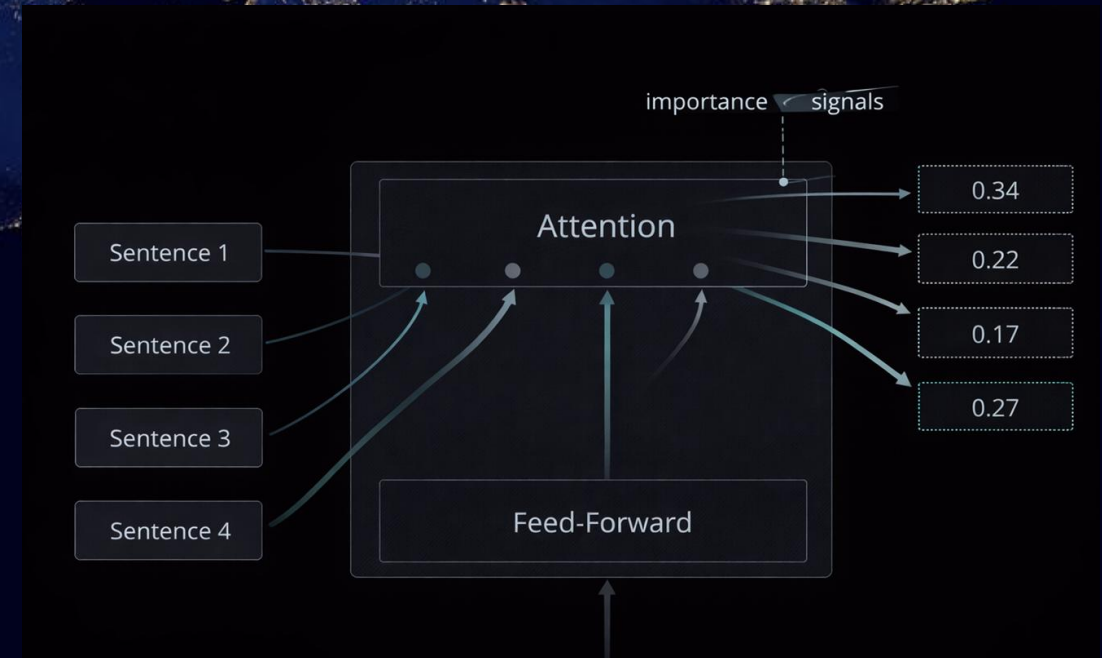


# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Transformer-Based Summarization

- Uses full-document context
- Learns semantic importance
- State-of-the-art performance
- Dominant in production systems

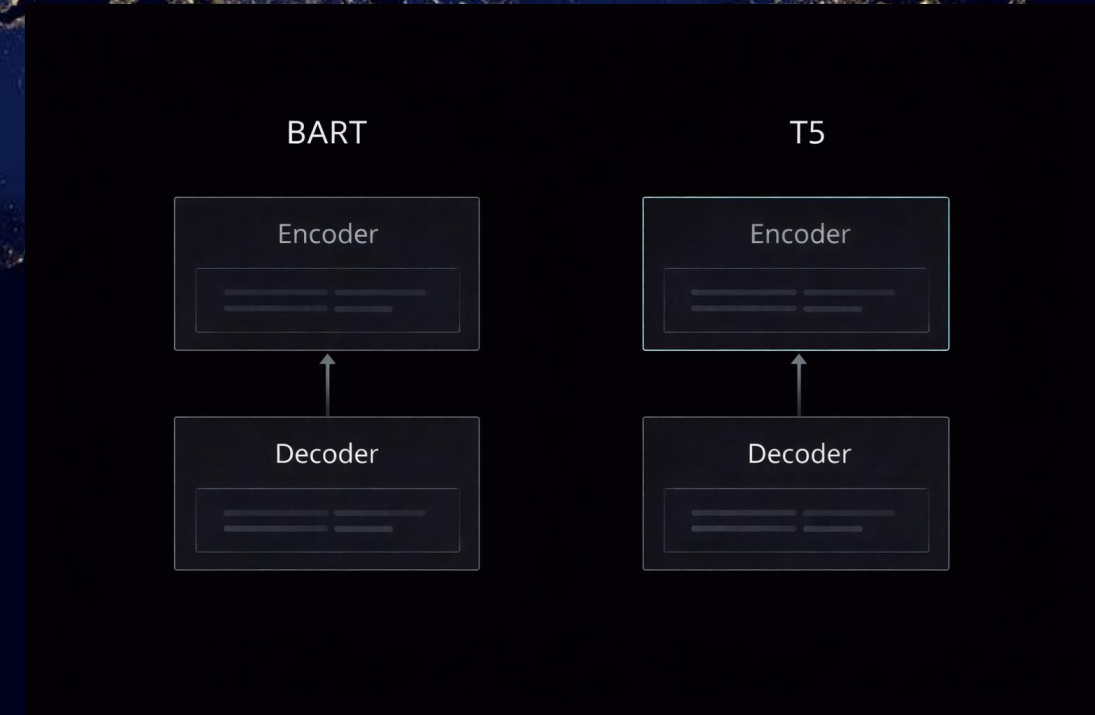


# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### BART and T5 Models

- Encoder–Decoder architectures
- Pre-trained for text reconstruction
- Strong performance on long documents
- Industry standard for summarization



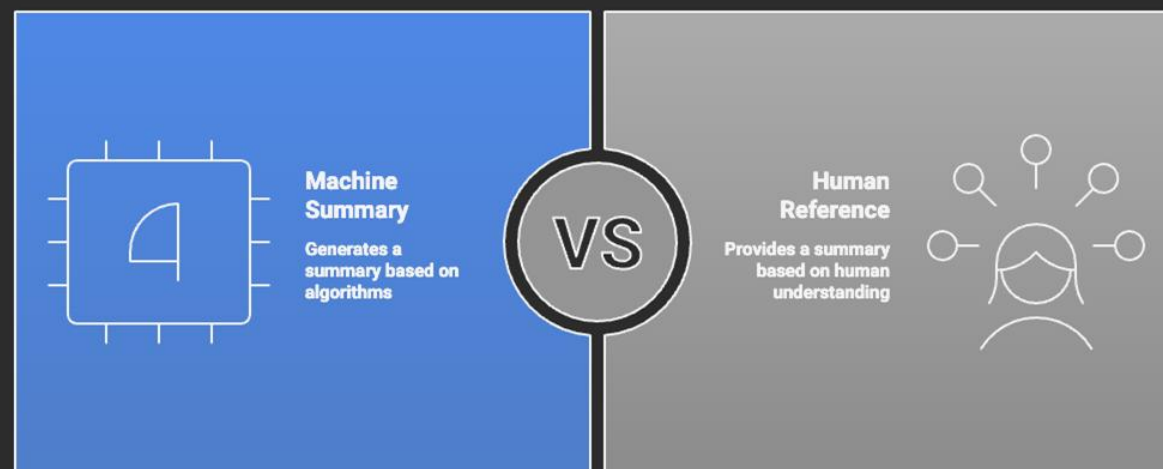
# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Evaluation Metrics (ROUGE)

- Measures overlap with human summaries
- ROUGE-1, ROUGE-2, ROUGE-L
- Focuses on recall
- Does not ensure factuality

#### How to evaluate machine-generated summaries?



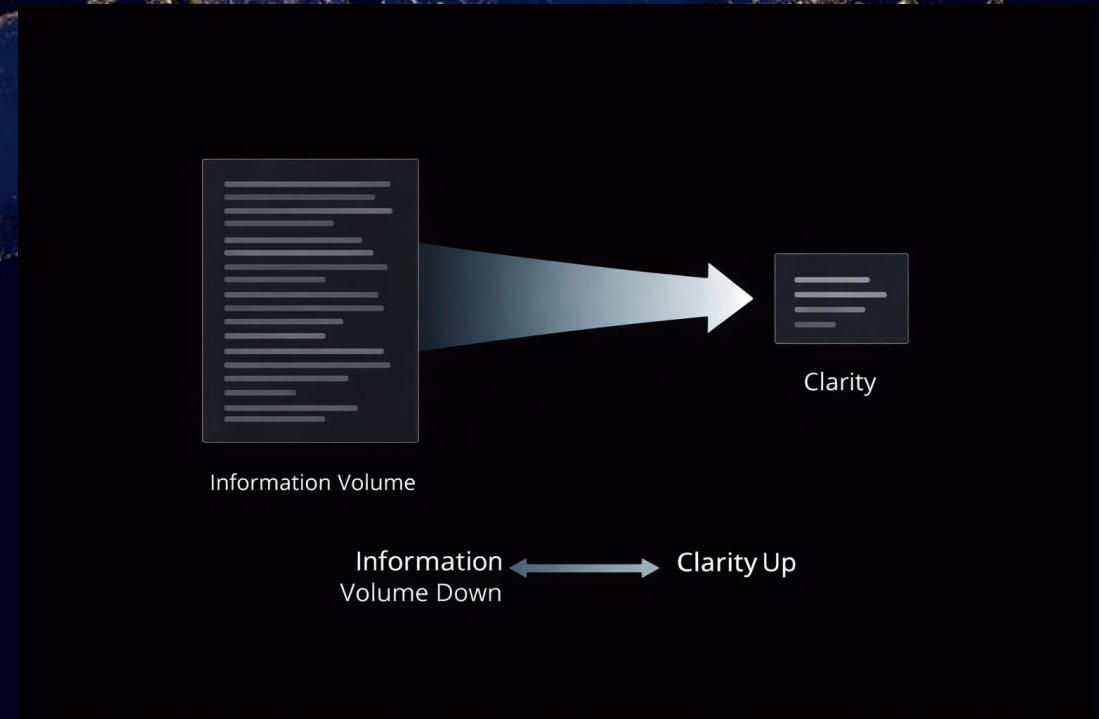


# M05:Applied NLP Tasks

## Chapter 5.4 Reducing Information Overload: Text Summarization

### Key Takeaways

- Summarization manages information overload
- Extractive = safer, abstractive = denser
- Transformers dominate modern systems





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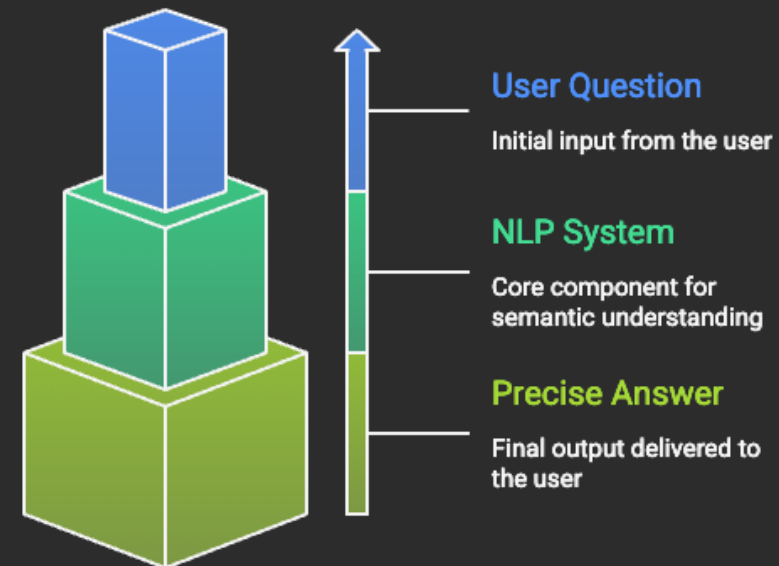
# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### What is Question Answering (QA)?

- Task: answer questions using text sources
- Goes beyond keyword search
- Requires language understanding
- Core of modern AI assistants

#### Enterprise AI QA Pipeline



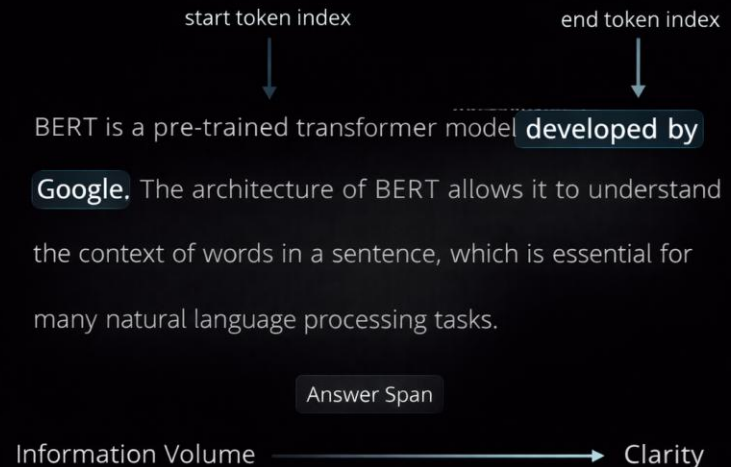


# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Extractive Question Answering

- Answer is a text span
- Start and end token prediction
- High factual precision
- Limited to provided context

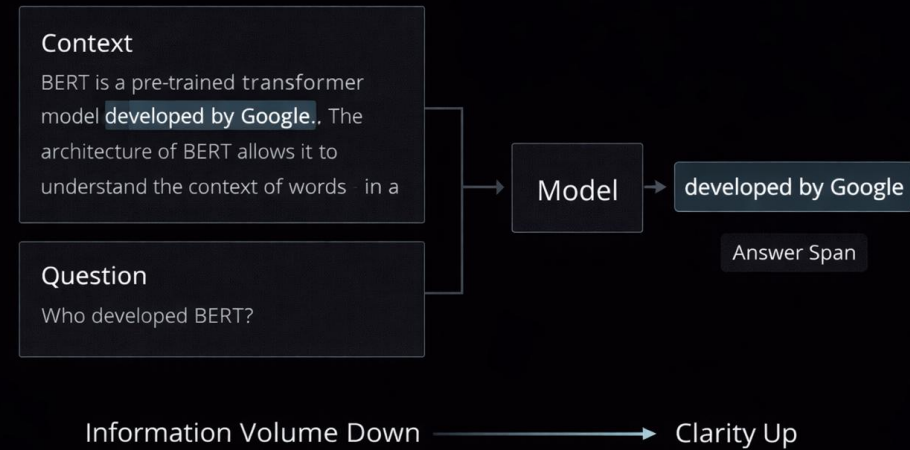


# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### The SQuAD Paradigm

- Benchmark dataset for extractive QA
- Context + Question → Span
- Enabled large-scale QA training
- Foundation for enterprise FAQ systems



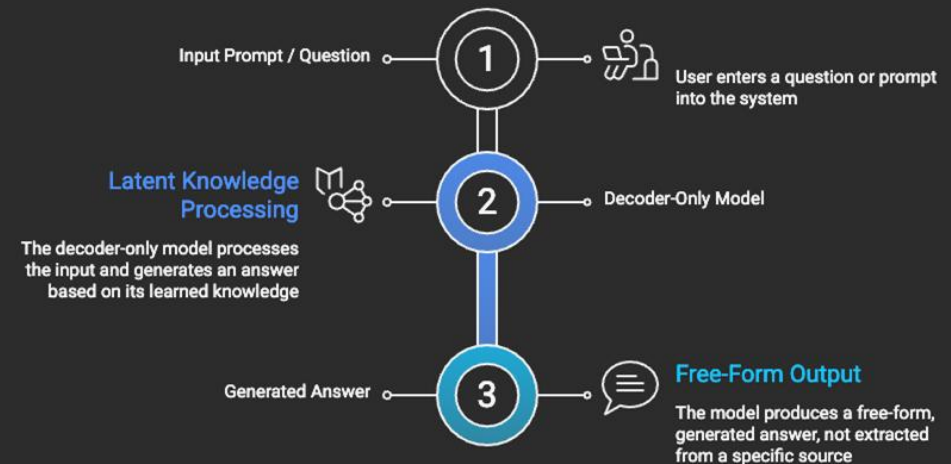
# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Generative Question Answering

- Benchmark dataset for extractive QA
- Context + Question → Span
- Enabled large-scale QA training
- Foundation for enterprise FAQ systems

#### Decoder-Only Model Answer Generation Process



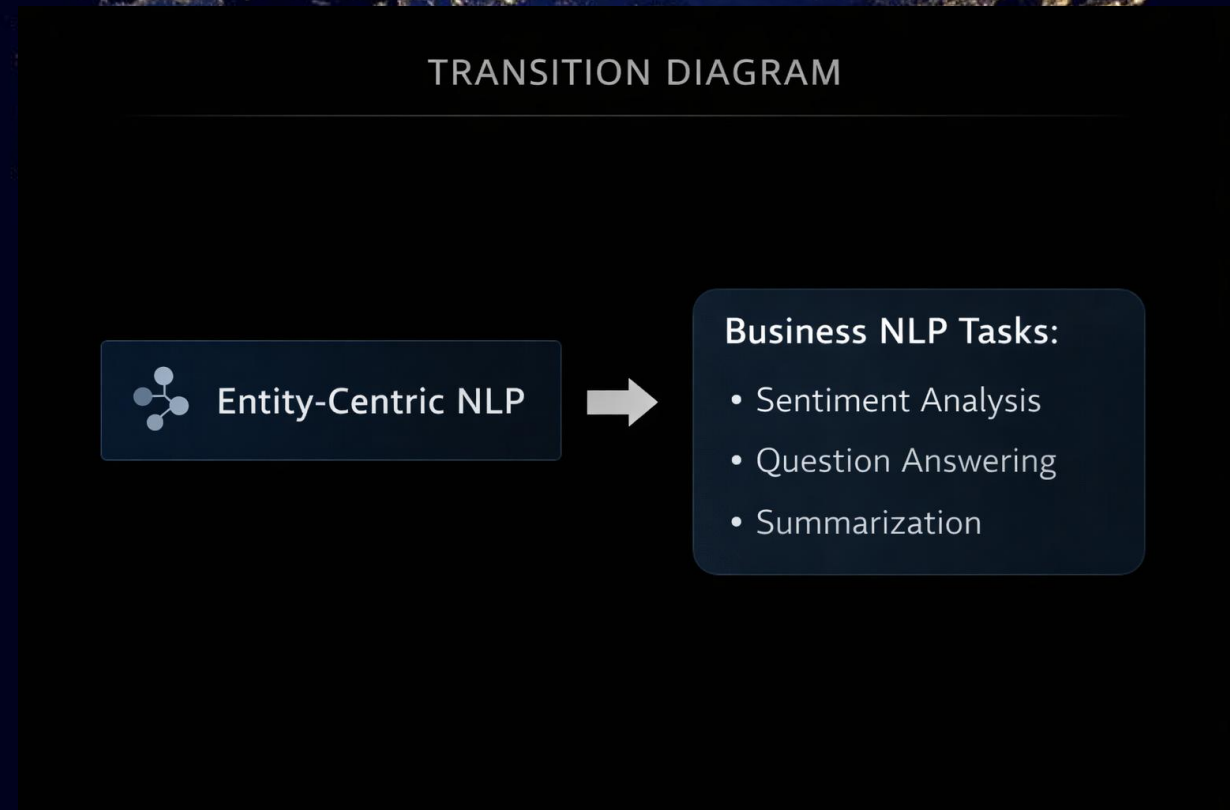


# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Knowledge-Based Question Answering

- Uses structured databases
- Converts language to queries
- Deterministic answers
- High reliability systems

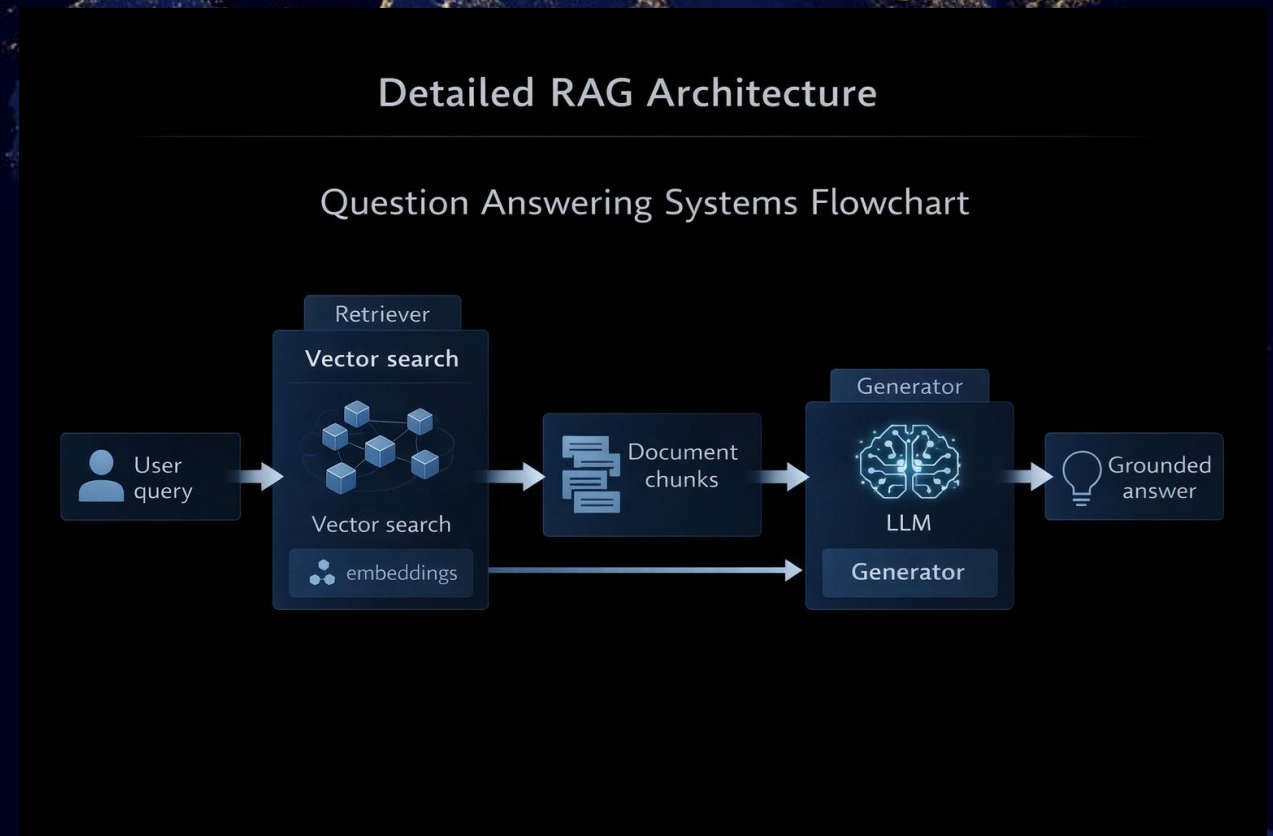


# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Retrieval-Augmented Generation (RAG)

- Industry-standard QA architecture
- Combines retrieval + generation
- Uses private documents
- Reduces hallucinations



# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Vector Databases in QA

Exact Match (EM)

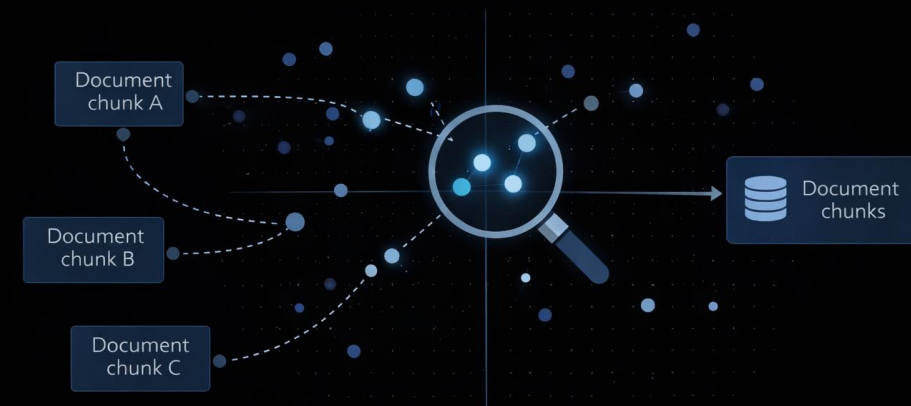
F1-Score (span overlap)

Faithfulness (RAG-specific)

Citation correctness

#### Technical Embedding Visualization

Embeddings stored in vector space with nearest-neighbor search retrieving document chunks



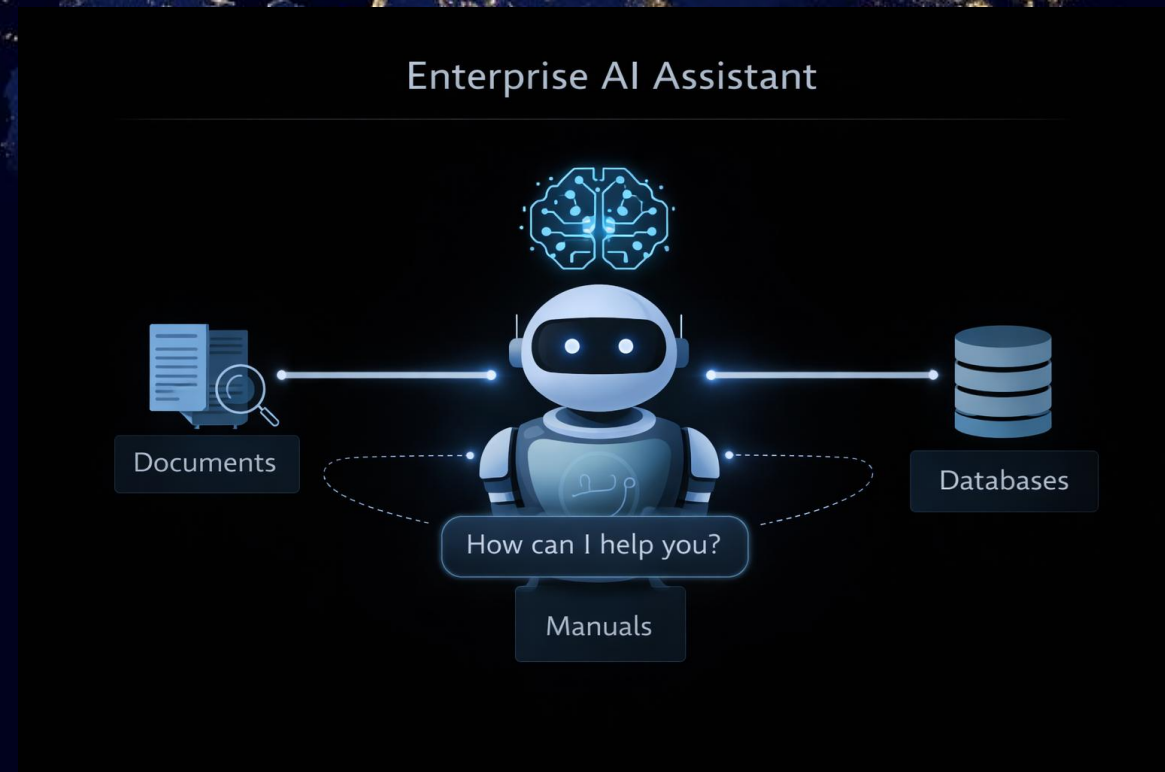


# M05:Applied NLP Tasks

## Chapter 5.5 Information Retrieval: Question Answering Systems

### Evaluation Metrics for QA

- Customer support automation
- Technical documentation search
- Legal and medical assistants
- Enterprise knowledge access



# M05:Applied NLP Tasks

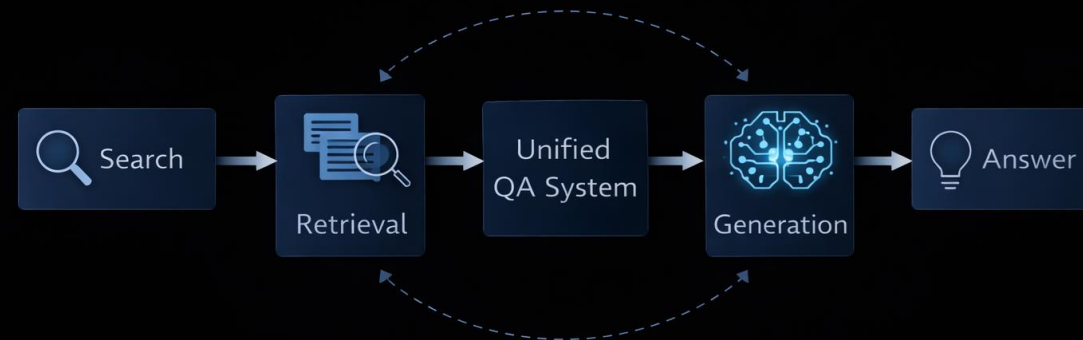
## Chapter 5.5 Information Retrieval: Question Answering Systems

### Key Takeaways

- QA bridges search and reasoning
- Extractive = precise, generative = flexible
- RAG is the dominant paradigm
- Grounding is critical for trust

### Closing Summary

Combining search, retrieval, and generation into a unified QA system





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5.6 Task Comparison and the Future of Applied NLP



# M05:Applied NLP Tasks

## 5.6 – Task Comparison and the Future of Applied NLP

### What is Question Answering?

Question Answering (QA) aims to provide direct answers to natural language questions by understanding both the query intent and the information source, going beyond traditional keyword-based search.

- From document retrieval to answer generation
- Requires semantic understanding
- Central to AI assistants and search systems
- Combines language and knowledge
- Foundation of enterprise AI

#### Evolution from Keyword Search to Question Answer



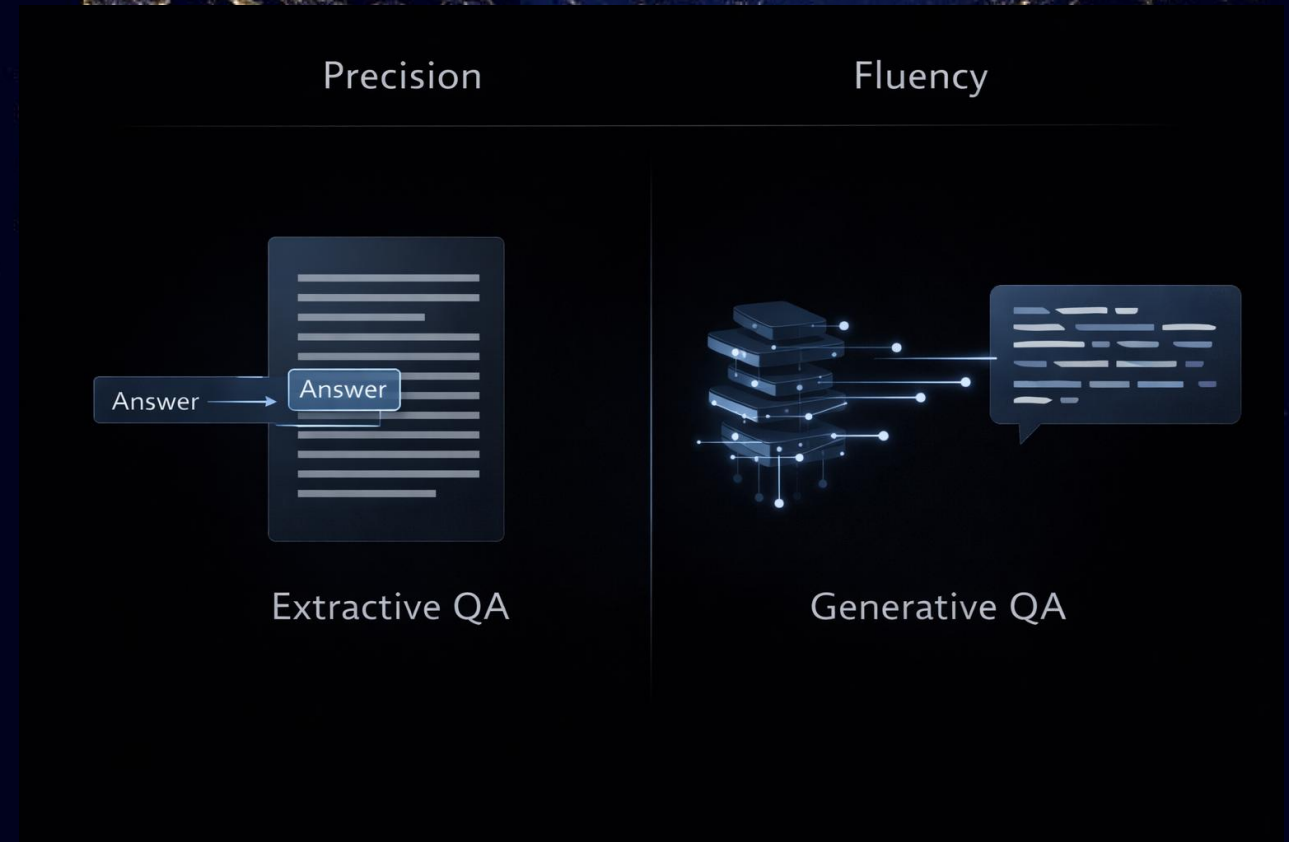
# M05:Applied NLP Tasks

## 5.6 – Task Comparison and the Future of Applied NLP

### Extractive vs Generative QA

QA systems operate under two main paradigms: extracting exact answers from text or generating new answers using language models.

- Extractive: answer is a text span
- High factual precision
- Used in legal and technical domains
- Generative: produces new text
- More fluent, higher hallucination risk



# M05:Applied NLP Tasks

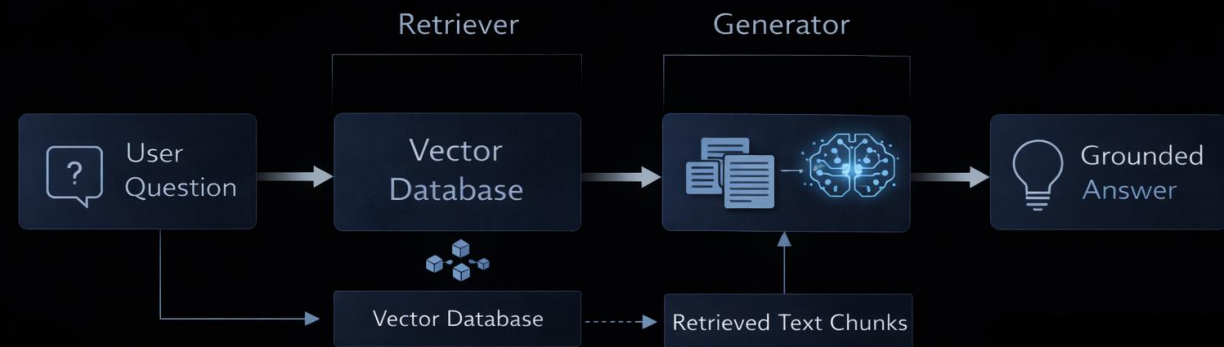
## 5.6 – Task Comparison and the Future of Applied NLP

### Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines document retrieval with generative models to produce answers grounded in real data.

Puntos clave:

- Industry standard for QA
- Uses vector search
- Injects retrieved text into prompts
- Reduces hallucinations
- Enables private data QA





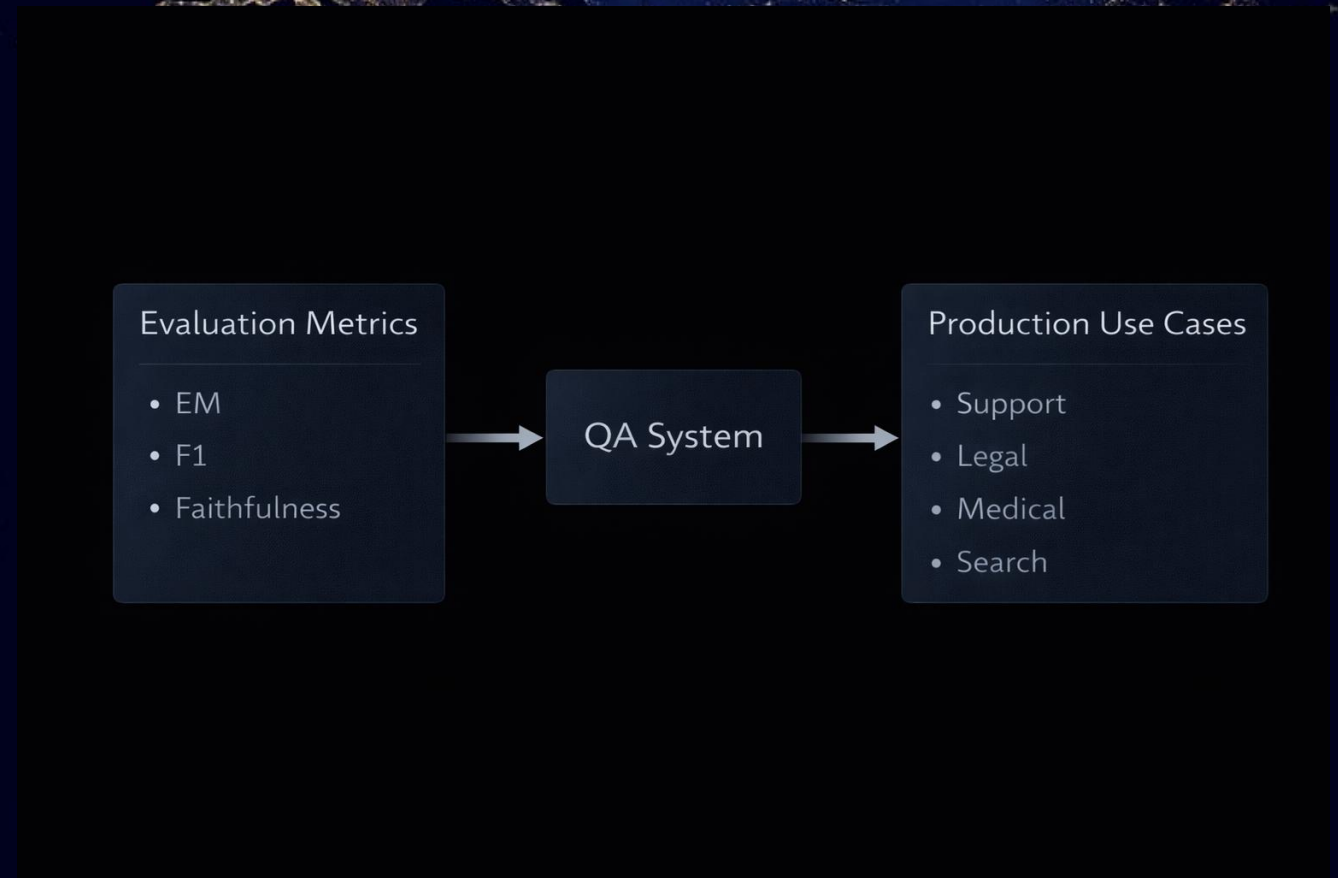
# M05:Applied NLP Tasks

## 5.6 – Task Comparison and the Future of Applied NLP

### Evaluation and Production Use

Reliable QA systems must be evaluated not only for correctness, but also for trust and factual grounding.

- Exact Match (EM)
- F1-score (span overlap)
- Faithfulness (RAG answers)
- Enterprise QA systems
- Trust and explainability



# M05:Applied NLP Tasks

## Resources & Academic References

### Core Benchmarks & Tasks

- Wang et al. (2018) — *GLUE: A Multi-Task Benchmark for Natural Language Understanding*  
<https://arxiv.org/abs/1804.07461>
- Wang et al. (2019) — *SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems*  
<https://arxiv.org/abs/1905.00537>
- Rajpurkar et al. (2016) — *SQuAD: 100,000+ Questions for Machine Comprehension of Text*  
<https://arxiv.org/abs/1606.05250>

### Text Classification & Zero-Shot Learning

- Cortes & Vapnik (1995) — *Support-Vector Networks*  
<https://link.springer.com/article/10.1007/BF00994018>
- Yin et al. (2019) — *Benchmarking Zero-shot Text Classification*  
<https://arxiv.org/abs/1909.00161>

### Sentiment Analysis

- Pang & Lee (2008) — *Opinion Mining and Sentiment Analysis*  
<https://www.cs.cornell.edu/home/llee/omsa/omsa.pdf>
- Pontiki et al. (2014) — *SemEval-2014 Task 4: Aspect-Based Sentiment Analysis*  
<https://aclanthology.org/S14-2004.pdf>

### Text Summarization

- Barzilay & Elhadad (1997) — *Using Lexical Chains for Text Summarization*  
<https://aclanthology.org/P97-1002.pdf>
- Lewis et al. (2019) — *BART: Denoising Sequence-to-Sequence Pre-training*  
<https://arxiv.org/abs/1910.13461>
- Raffel et al. (2019) — *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (T5)*  
<https://arxiv.org/abs/1910.10683>

### Question Answering & RAG

- Lewis et al. (2020) — *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*  
<https://arxiv.org/abs/2005.11401>

### Ethics & Future of Applied NLP

- Bender et al. (2021) — *On the Dangers of Stochastic Parrots*  
<https://dl.acm.org/doi/pdf/10.1145/3442188.3445922>
- Bommasani et al. (2021) — *On the Opportunities and Risks of Foundation Models*  
<https://arxiv.org/abs/2108.07258>